

WASP-50b

R-1318

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_201229 · created 2026-07-31T19:21:26+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459212.696 +/- 0.042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.05
Transit depth (Rp/Rs)^2	3.02 +/- 1.74 [%]
Orbital Inclination (inc)	88.72 +/- 2.91
Airmass coefficient 1 (a1)	0.64 +/- 0.31
Airmass coefficient 2 (a2)	0.24 +/- 0.49
Scatter in the residuals of the lightcurve fit is	47.25 %
Best Comparison Star	None
Optimal Aperture	4.51
Optimal Annulus	38.93
Transit Duration (day)	0.078 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.05 vs 0.1404 (deviation 0.61σ)
Duration fit vs published	0.078 d vs 0.0754 d (deviation 0.11σ)
Mid-time O–C	21.2 min
Depth SNR	3.1
Residual scatter	47.25 %

Light curve

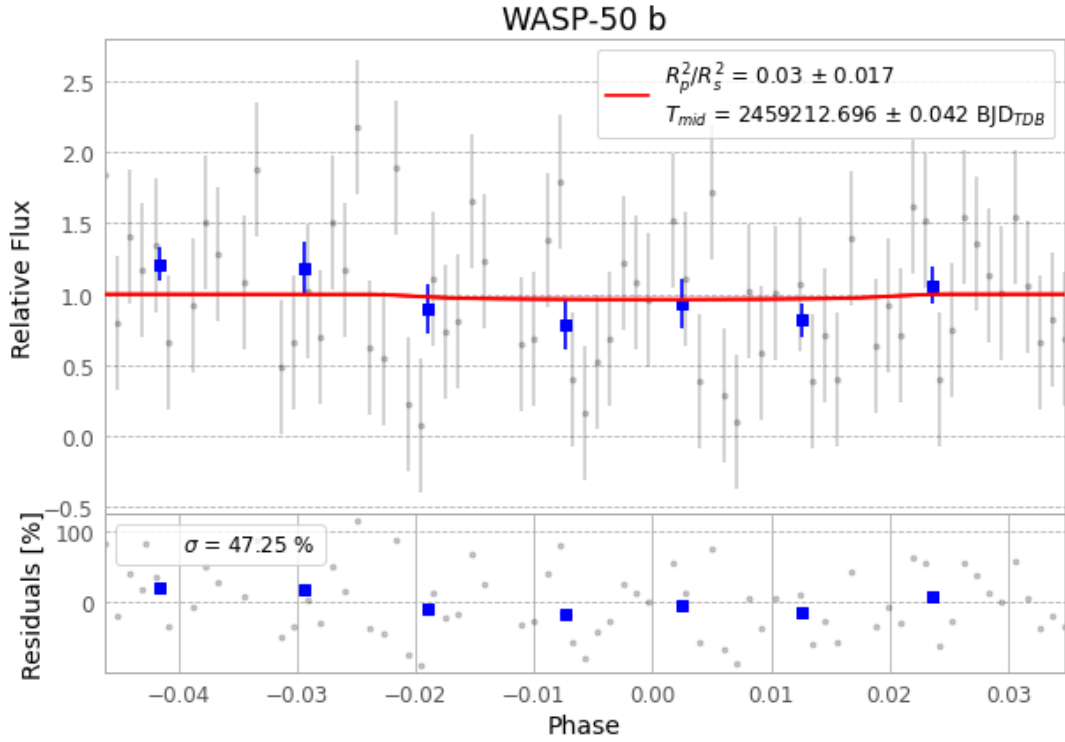


Figure 1 — FinalLightCurve_WASP-50 b_2020-12-28.png (SHA-256 01be3e7b0ad7e23a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [585, 270], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2adbd1b4ef885d7a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-50201229022712.FITS → WASP-50201229061512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-201229.log (015d2abcad15...), FinalLightCurve_WASP-50 b_2020-12-28.png (01be3e7b0ad7...), FinalLightCurve_WASP-50 b_2020-12-28.csv (f1415ce847b6...)

Compute resources

Wall time 365.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 19:21Z.